＊研究論文＊

Iterative Region-of-Interest Reconstruction From Truncated CT Projection Data Under Blind Object Support

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Abstract

We have investigated image reconstruction from truncated projection data in computed tomography. Region-of-interest (ROI) imaging techniques, which are practically implemented by limiting X-ray exposure to the target region of the object, provide so-called truncated projection data. This limitation in the projection data converts the image reconstruction problem to an undetermined problem in which extra effort is needed to achieve accurate ROI reconstruction. Recently developed theories based on differentiated backprojection (DBP) suggest the possibility of achieving accurate ROI imaging in some clinical imaging scenarios. An essential requirement for achieving exact ROI reconstruction is prior knowledge of the object support (OS). However, the exact OS is not always available in many imaging applications. In this paper, first, we investigate how the image artifacts depend on the accuracy of the *a priori* known OS, and second, we propose an EM-based iterative reconstruction algorithm to reduce the ROI artifacts when the OS is unknown. The image reconstruction cost function is modified to include a thresholding function in the form of ℓ_0 norm, and thanks to this modification, the OS is automatically detected by the thresholding function and the ROI artifacts are reduced. The experimental results, including numerical simulations and real data with different ROI and object configurations, indicate that a significant reduction in ROI artifacts can be achieved by using the proposed algorithm.

Key words: Tomography, Image reconstruction, ROI imaging, Truncated projection data, Object support

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1. Introduction

Computed Tomography (CT) has become the most popular medical imaging modality to diagnose various diseases. The widespread use of CT increased the radiation dose in the population as CT involves much higher dose over the traditional plain-film radiography〔1〕. The interest was recently focused on developing reconstruction algorithms from low dose CT projection data to reduce the radiation dose. The dose reduction is challenging as the decreasing of the x-ray beam intensity increases the statistical noise in the reconstructed images. One possible approach to achieve the dose reduction is the Region-of-Interest (ROI) CT imaging by focusing the radiation only to the organ of interest, which reduces unnecessary radiation to the remaining portion of the patient. This imaging setup directly leads to data truncation problem in which left and/or right portions of the projection data are missing for all the view angles. **Fig. 1** illustrates CT imaging setups for the full scan and the ROI scan. The topic of ROI imaging has been studied for a long time (e.g.〔2 ~ 5〕) and, based on the concept of analytical reconstruction, it had been believed that ROI reconstruction, even for a small ROI, requires all projection rays passing through the whole object and any truncation leads to losing the solution exactness. However, the recently developed theories based on

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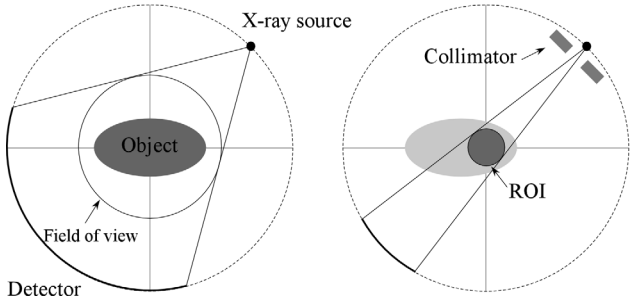

Fig. 1 CT imaging configurations for whole object imaging (left) and ROI imaging (right).

the concept of Differentiated Backprojection (DBP)〔6〕and Backprojection-Filtration (BPF)〔7〕succeeded in reducing the required set of projection rays for theoretically exact and stable ROI reconstruction. It became clear that the position of the ROI is an important factor to determine the possibility of accurate ROI reconstruction. Defrise et al.〔8〕provided a sufficient condition for accurate ROI reconstruction with truncated projection data. The exact reconstruction requires that the ROI contains a small limited region inside〔9, 10〕or outside the object with known intensity values. Practically, the availability of accurate prior information inside the object is difficult, however selecting this region as a portion of the background is more feasible as the attenuation value of air is known to be almost zero. Enforcing this *a priori* information of the background region is usually done using the Object Support (OS) constraint during image reconstruction. The OS is the region where the object is certainly inside. In real applications, however, the *a priori* knowledge of the OS is not always available or requires extra efforts to be estimated accurately. Alternatively, OS is selected a slightly larger than the true object to ensure that the object is located completely inside. Through numerical studies, it is known that the lack of exact OS produces DC-shift and low-frequency artifacts in the reconstructed images〔11, 12〕.

The aim of this paper is two-fold. The first aim is to evaluate the relation between the accuracy of prior knowledge of OS constraint and the quality of image reconstructed from truncated projection data. The second aim is to propose an iterative ROI reconstruction algorithm from truncated projection data when the OS is unknown or technically unavailable. Throughout this paper, we consider the truncated projection data such that only the projection rays passing through the ROI are measured. These settings are suitable to achieve a reduction in the radiation exposure outside the ROI. In the proposed algorithm, the cost function is modified to include a thresholding function in the form of ℓ_0 norm distance. The OS is then automatically estimated by the thresholding function and iteratively converges towards the exact one. In section 2, we explain necessary basic concepts and the proposed iterative reconstruction algorithm. In section 3, we present numerical results of simulation studies and real data. In section 4, we conclude the paper.

2. Methodology

1）Object support constraint

We assume that the projection data is truncated such that it includes only the projection rays corresponding to all lines passing through a limited ROI Ω. We represent the OS (the *a priori* known region where the object is certainly inside) by μ. Then, the exact reconstruction of Ω is possible if there exists a set of pixels x_j such that $x_j \in \Omega$ and $x_j \notin \mu$〔8〕. In other words, ROI should be defined such that it includes a limited region outside the OS where the intensity value is *a priori* known to be zero. Two different imaging configurations are shown in **Fig. 2** that illustrates Defrise's condition. In the setup of (a), the exact reconstruction is possible as the ROI is selected such that it contains a small region outside the OS μ (hatched area), while the exact reconstruction is theoretically not possible in the setup of (b) as it corresponds to the so-called *interior problem*〔13〕. The challenging of this theory is that, when the accuracy of OS is worse, the observed ROI should be selected to cover a larger region of the scanned

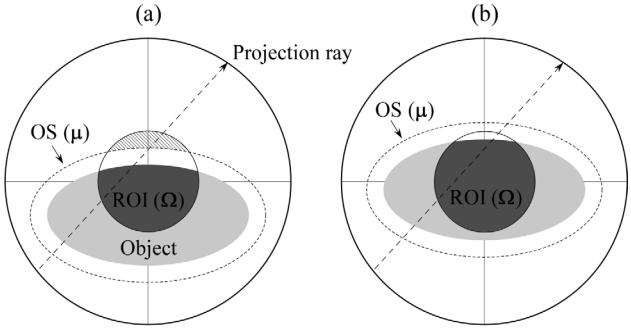

Fig. 2 Two different ROI imaging configurations, (a) ROI contains a region (shaded) outside the OS (Defrise's condition is satisfied and the solution exactness is assured) and (b) ROI is contained completely inside the OS (interior problem and the solution is not unique).

object and a larger area outside the object. A typical clinical application that demonstrates these imaging configurations is the cardiac ROI imaging. To obtain a theoretically accurate reconstruction, the ROI should be selected carefully such that it includes a region where the intensity value is *a priori* known to be zero. Considering the fact that it is difficult to determine the exact contours for different patients (exact OS) in prior to imaging, the ROI should be selected large enough to include the target object (heart) and the region outside the OS similarly to the setup of **Fig. 2**(a). The aim of this work is to estimate the exact OS during image reconstruction. This will allow us to reduce the size of ROI leading to dose reduction and other benefits. Finally, we remark that analytical solution have not been found yet for this truncation problem so that using iterative algorithms is an only way to perform image reconstruction in this problem〔8~10〕.

2）Iterative algorithms with zero-pixels identification

To estimate the OS, for example, iterative algorithms which alternate image reconstruction and zero-pixels identification (by thresholding or image segmentation) can be considered. However, such brute-force algorithms cannot be implemented efficiently and their convergences are not guaranteed in general. In this paper, we introduce a more sophisticated class of iterative algorithms based on the relatively new concept of so-called sparse signal recovery or compressed sensing〔14〕. The main idea is to include a distance function consisting of ℓ_1/ℓ_0 norm of reconstructed images into the cost function for image reconstruction. Then, this additional ℓ_1/ℓ_0 norm term works as a thresholding operation applied after each iteration step, which replaces small pixels values to zeros (i.e. identifying zero-pixels). Here, the word "zero-pixels" refers to pixels which have zero pixel values. As a result, unlike the ordinary iterative algorithms such as ART and EM algorithm, this new class of iterative algorithms can automatically estimate the OS during image reconstruction. When this concept is combined with the EM iterative reconstruction algorithm, it was shown that the resulting algorithm takes the form of alternating the following two substeps〔15〕. The first substep is the regular iteration step of EM algorithm, and the second substep is the thresholding operation called the hard/soft thresholding in the wavelet literature. The accuracy of zero-pixels identification is improved as the iteration proceeds, and the estimated OS approaches to the exact object boundary.

Recently, such algorithms have been investigated to solve general inverse problems, and are attracting attention of many researchers (e.g.〔16, 17〕). Our group also applied this class of iterative algorithms to 3D blood vessel reconstruction from limited number of projections〔18〕and PET/SPECT reconstruction〔15〕. In this paper, we use the same concept to estimate the OS in image reconstruction from truncated projection data. As shown later, this concept leads to automatically estimating the OS during the iteration and thereby reducing the DC-shift and low-frequency artifacts significantly.

3）Proposed reconstruction algorithm

In this section, we explain the proposed reconstruction algorithm. First, we note that the iterative algorithm here is essentially same as that in〔15〕, though the target imaging application and the nature of projection data in this

paper are completely different from that in 〔15〕. Let $\vec{x}=(x_1,...,x_n)$ denote the image vector and $\vec{y}=(y_1,...,y_m)$ denote the measured projection data. These two vectors are related to each other by the linear equation $A\vec{x}=\vec{y}$, where $A=\{a_{ij}\}$ is the $m\times n$ system matrix. Then, the proposed cost function for image reconstruction is expressed as

$$f(\vec{x})=L(\vec{x})+\beta T(\vec{x}), \tag{1}$$

Where $L(\vec{x})$ is the negative log-likelihood function, $T(\vec{x})$ is the distance function between the reconstructed image $\vec{x}$ and the null image, and β is the hyperparameter. The definition of $L(\vec{x})$ and $T(\vec{x})$ are given by

$$L(x)=\sum_{i=1}^{m}\left[\sum_{j=1}^{n}a_{ij}x_j-y_i\log\left(\sum_{j=1}^{n}a_{ij}x_j\right)\right] \tag{2}$$

$$T(\vec{x})=\sum_{j=1}^{n}s(x_j),\ \text{ and }\ s(t)=\lim_{\varepsilon\to+0}|t|^{\varepsilon}\equiv\begin{cases}1 & t\neq 0\\ 0 & t=0\end{cases} \tag{3}$$

We derive the iterative reconstruction algorithm by using the majorization-minimization strategy same as that in 〔15〕. The essential outline of the derivation is as follows. At each iteration k, we approximately majorize the non-separable part of the cost function $L(\vec{x})$ by a separable quadratic function $L(\vec{x};\vec{x}^{(k)})$ around $\vec{x}=\vec{x}^{(k)}$ as:

$$\begin{aligned}L(\vec{x};\vec{x}^{(k)}) &= L(\vec{x}^{(k)})+\nabla L(\vec{x})\,|_{\vec{x}=\vec{x}^{(k)}}\,(\vec{x}-\vec{x}^{(k)})^T+\frac{1}{2}\sum_{j=1}^{n}\frac{\sum_{i=1}^{m}a_{ij}}{x_j^{(k)}}(x_j-x_j^{(k)})^2\\ &= \sum_{j=1}^{n}\frac{\sum_{i=1}^{m}a_{ij}}{2x_j^{(k)}}(x_j-p_j)^2+C(\vec{x}^{(k)})\end{aligned} \tag{4}$$

$$p_j=\frac{x_j^{(k)}}{\sum_{i=1}^{m}a_{ij}}\sum_{i=1}^{m}\frac{a_{ij}y_i}{\sum_{j'=1}^{n}a_{ij'}x_{j'}^{(k)}}\ , \tag{5}$$

where $C(\vec{x}^{(k)})$ is the term independent of $\vec{x}$. Consequently, the final form of the separable surrogate cost function is given by

$$f(\vec{x};\vec{x}^{(k)})=\sum_{j=1}^{n}\beta[t_j(x_j-p_j)^2+s(x_j)]+C(\vec{x}^{(k)})\ \text{ with }\ t_j=\frac{1}{2\beta x_j^{(k)}}\sum_{i=1}^{m}a_{ij} \tag{6}$$

At each iteration number k, the above cost function $f(\vec{x};\vec{x}^{(k)})$ is minimized instead of $f(\vec{x})$ to obtain the next iterate $\vec{x}^{(k+1)}$. Setting the partial derivative of Eq. (6) with respect to $\vec{x}$ equal to zero and then applying the non-negativity constraint $\vec{x}\geq 0$ yields the iteration formula given below by Eq. (7). The proposed iterative algorithm, which we call Th-OSEM (Threshold ordered subsets expectation maximization), is summarized as follows:

[STEP 1] (Initialization) Set the initial image as $\vec{x}^{(0)}=\varepsilon$ where $\varepsilon>0$ is a small positive number. Set the iteration number as $k\leftarrow 0$.

[STEP 2-1] (Majorization) Around the current iterate $\vec{x}^{(k)}$, $L(\vec{x})$ is (approximately) majorized using the separable function of Eq. (4) to get the surrogate cost function of Eq. (6).

[STEP 2-2] (Minimization) We minimize the cost function of Eq. (6) over $\vec{x}\geq 0$ to obtain the next iteration $\vec{x}^{(k+1)}$:

$$\vec{q}=\arg\min_{\vec{x}} f(\vec{x},\vec{x}^{(k)}),\ \ \vec{x}_j^{(k+1)}=\max(q_j,\delta) \tag{7}$$

where $\delta>0$ is a small value to guarantee that $x^{(k+1)}>0$.

[STEP 3] (Convergence check) Increment the iteration number as $k\leftarrow k+1$ and go to **[STEP 2-1]** until reaching to a stopping criterion.

The practical implementation of the proposed algorithm is simply described as follows. In **[STEP 2-1]**, the majorization substep is applied by computing $\vec{p}=(p_1,p_2,...,p_n)$ using Eq. (5) which is same as executing a single

iteration of the conventional OSEM algorithm. The minimization in **[STEP 2-2]** can be explicitly performed by the following thresholding function〔15〕.

$$q_j = \begin{cases} 0 & p_j \le \sqrt{1/t_j} \\ p_j & elsewhere \end{cases}, \tag{8}$$

The Th-OSEM algorithm is executed as one substep of the conventional OSEM algorithm (Eq. (5)) followed by the pixel-by-pixel thresholding (Eq. (8)). The OSEM algorithm was used as a basic iterative algorithm to develop the proposed iterative method, because the non-negativity feature of the EM-based algorithms is helpful in regularizing the solution and estimating the OS more accurately. However, we note that the similar algorithms can be developed based on the other iterative algorithms such as ART and the conjugate gradient. The thresholding function in Eq. (8) has a large similarity to the well-known hard-thresholding operation in the wavelet denoising. This proposed algorithm belongs to a class of recently developed iterative reconstruction algorithms which use a sparsity constraint by employing a distance function based on the ℓ_1/ℓ_0 norm〔15 ~ 18〕. A useful theoretical analysis of convergence can be found in〔19〕.

4）Handling the hyperparameter β

In maximum *a posteriori* (MAP) reconstruction algorithms, the hyperparameter β in Eq. (1) is known as the regularization parameter which handles the strength of the prior term. In the proposed algorithm, β controls the size of the thresholding window as in Eq. (8). It is clear that β should be carefully selected, because, if it is assigned to a relatively large value, some low-frequency details in the reconstructed image will be lost due to the strong thresholding. In contrast, if β is relatively small, the merit of thresholding is small and the reconstruction algorithm behaves similar to the conventional iterative algorithms. We propose to use a dynamic value for this parameter by starting with a relatively large value (β_0) and then gradually decreasing it to zero according to the following rule.

$$\lim_{k\to\infty} \beta_k = 0 \quad \text{and} \quad \sum_{k=0}^{\infty} \beta_k = \infty \tag{9}$$

A simple setup to satisfy Eq. (9) is to use $\beta_k = \beta_0/(1+k)$. This approach is based on the observation that, in the early iterations, a strong thresholding is required for the rough detection of the OS and a weak thresholding is preferred later to retrieve missing image details. We expect that the iteration converges to an approximate solution to the following optimization problem.

$$\text{Minimize} \quad T(\vec{x}) \quad \text{subject to} \quad A\vec{x} = \vec{y},\ \vec{x} \ge 0 \tag{10}$$

The rationale behind why the dynamic value of hyperparameter β_k should satisfy Eq. (9) is as follows. First, the left equation in Eq. (9) is necessary, because the value of β in the cost function of Eq. (1) should be close to zero for the solution minimizing $f(\vec{x})$ to coincide with the solution to the constrained problem of Eq. (10). On the other hand, the right equation in Eq. (9) intuitively means that the speed to decrease the value of β_k to zero should be sufficiently slow. This constraint is necessary, because, if the speed to decrease the value of β_k is too fast, the iteration is trapped at some point before reaching to the solution minimizing $T(\vec{x})$. For example, $\beta_k = \beta_0/(1+k)$ satisfies this constraint, but $\beta_k = \beta_0/(1+k)^2$ does not. We also note that the similar constraints have been used in the row-action maximum likelihood algorithm (RAMLA) for image reconstruction in emission tomography〔20〕and the gradient descent method with diminishing stepsize〔21, p51〕.

3. Numerical results

1）The effect of OS

Simulation studies were performed to evaluate the effect of *a priori* knowledge of the OS to the quality of reconstructed images. These studies were performed using a single slice (z = 0) of the FORBILD thorax phantom with image matrix size of 512×512 pixels. The ROI was set to be a circle with a diameter of 160 pixels covering the cardiac region as shown in **Fig. 3**(a). The ROI position was selected such that the condition of Defrise is satisfied when the OS is close to the exact one (i.e. the ROI contains a small region outside the exact OS). The projection data were

computed with parallel-beam geometry for 512 radial bins and 512 views (over 180°), and the data were truncated manually such that only projection rays passing through the selected ROI are used for the reconstruction. Different selections of the OS constraint μ were applied that include exact OS knowledge, the case where the OS is known with different degrees of OS accuracy (110% ~ 150% larger than the exact OS) and the unknown OS. The "exact OS" means the true exact contour of the scanned object. The "110% OS" means the region inside the ellipse whose semi-axis length is 10% larger than that of the ellipse corresponding to the "exact OS". The definitions of "130% OS" and "150% OS" are similar (see **Fig. 3**(a)). The reconstruction was performed using the conventional OSEM algorithm with 8 subsets and 100 iterations for each setup. The implementation of the iterative reconstruction is as follows:

[Step 1] The image matrix $\vec{x}$ in the iterative algorithm is prepared such that it contains the whole object (not the ROI $\mathbf{\Omega}$ only) though only the ROI is the target to be reconstructed accurately.

[Step 2] The whole image matrix $\vec{x}$ is reconstructed by OSEM (section 3.1) or Th-OSEM (section 3.2) using only a set of projection data $\vec{y}$ which intersect the ROI $\mathbf{\Omega}$. When the OS is *a priori* known, after every iteration k, pixel values outside the OS are set to zeros.

[Step 3] Trim the reconstructed image matrix $\vec{x}$ to the small image which contains only the ROI $\mathbf{\Omega}$.

Reconstructed ROIs are shown in **Fig. 3** and the corresponding ROI central horizontal and vertical profiles are shown in **Fig. 4**. In **Fig. 3**, the reconstructed whole image matrix (512×512 pixels) was trimmed to the ROI image consisting of 512×300 pixels for display. It can be observed that improving the OS accuracy results in reduction of the DC-shift and low-frequency artifacts in the reconstructed images.

We note that, in the problem dealt with in this paper, 100 iterations with 8 subsets are reasonable choices, which were determined from the experimental optimization of iteration number. The explanation is as follows. The OSEM algorithm is normally used in PET/SPECT reconstruction. Though 100 iterations with 8 subsets are too many in PET/SPECT applications, the nature of the reconstruction problem in this paper differs from that in PET/SPECT with respect to the following point. In CT, unlike PET/SPECT imaging, the object normally contains low-contrast structures and the compressed gray scale (e.g. [0.94, 1.03] in **Fig. 3**) is used for image display to make the low-contrast structures visible. To recover the low-contrast structures with allowable small DC/low-contrast artifacts, more numbers of iterations are necessary compared to the case of PET/SPECT. So, we have carefully optimized the iteration number in the experimental study. In **Fig. 5**, we showed the profiles of the reconstructed images with 10, 20, 50,100, and 1000 iterations of the OSEM algorithm with 8 subsets. The results demonstrate that the DC/low-contrast artifacts still continue to decrease up to approximately 100 iterations, but only very small improvements were

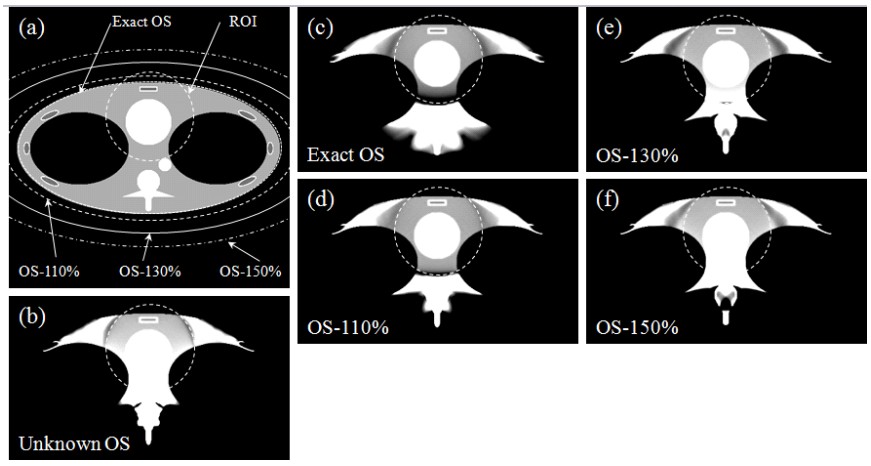

Fig. 3 (a) FORBILD thorax phantom (z = 0) and different OS setups, ROI reconstruction results with different OS degrees of accuracy, (b) OS is unknown, (c) OS is exact and (d) ~ (f) OS is known as the region that is larger than the exact OS. The display gray scale is [0.94, 1.03].

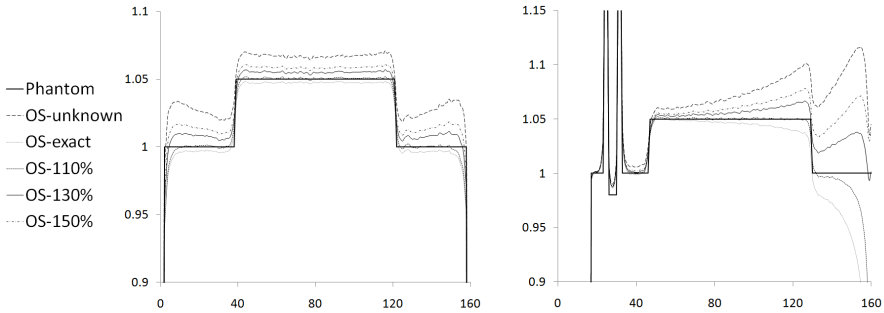

Fig. 4 Central horizontal (left) and vertical (right) profiles of ROI images shown in Fig. 3.

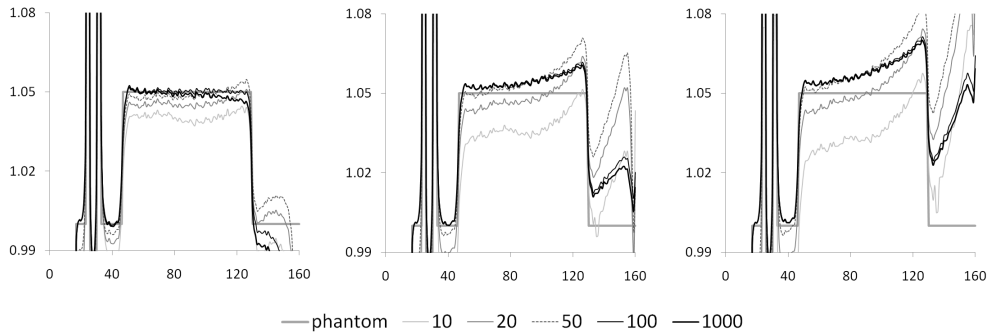

Fig. 5 Central vertical profiles of the reconstructed ROI images with different degrees of OS accuracy corresponding to 110% (left), 130% (middle) and 150% (right) using 10, 20, 50, 100, and 1000 iterations of OSEM algorithm with 8 subsets.

observed if we iterate more, from which we found that 10 or 20 iterations are not enough in the current problem. We also found that the necessary iteration number to achieve sufficient image quality did not depend very much on the OS accuracy as long as the OS is changed from "exact" to "150% OS". Furthermore, the convergence speed of the proposed algorithm with automatic OS estimation (detailed below in section 3.2) was only a bit slower compared to the cases where the OS is *a priori* known. From this experimental study (to determine the necessary iteration number) and the fact that the iterative algorithm with more than 100 iterations is not practical anymore, we performed all the simulation studies with 100 iterations.

2）Proposed iterative algorithm study

The ordered subsets version of the proposed algorithm, detailed above in section 2.3, was evaluated in comparison to the conventional OSEM algorithm. In this study, the FORBILD thorax phantom was modified by adding low-contrast hot/cold spots in the cardiac region to evaluate the effect of low-frequency artifacts. Other simulation parameters were similar to that in the previous study. The following three different scenarios were investigated. The first one assumes that the OS is exactly known (the ideal case) and reconstruction is done using the conventional OSEM algorithm. In the second and third scenarios, reconstruction is done using the OSEM and Th-OSEM with no *a priori* knowledge of the OS, respectively. Reconstructed ROIs for the noise free data and the data with added Poisson noise are shown in **Fig. 6** and the corresponding profiles are plotted in **Fig. 7**. The same image trimming as that in **Fig. 3** was used for display. It is clear that the proposed algorithm successfully reduces the DC-shift and low-frequency artifacts when the OS is unknown thanks to the automatic estimation of the OS.

From the reconstructed images in **Fig. 6** and the corresponding profiles in **Fig. 7**, it is noticed that the upper part of the ROI (close to the object boundary) suffers from less DC/low-frequency artifacts than the lower part (located at the center of object). In the followings we explain this phenomenon. It is known that the stability of inversion for

this reconstruction problem depends on the position of pixel inside the ROI. The inversion is more stable for the pixels located close to the upper part of the ROI (close to the object boundary), but the stability of inversion becomes unstable for the pixels located at the lower part (located at the center of object). This fact has been theoretically clarified by using the analytical continuation technique in〔8〕(please see **Fig. 5** of reference〔8〕). Here, we give an intuitive explanation for this variation in the stability from the view point of iterative reconstruction. For the pixels located at the upper part of the ROI, the boundary of OS is close so that the *a priori* knowledge that "the object is zero outside the OS" is quickly propagated to the pixels during iterative reconstruction, which result in better stability of inversion. However, for the pixels located at the lower part of the ROI, the closest OS is distant so that further iterations are necessary for the *a priori* knowledge to propagate to the reconstructed pixels, which result in worse stability.

From practical point-of-view, it is possible to correctly estimate a portion of the OS (corresponding to the top side of the object in the previous simulation study) from the truncated projection data directly by detecting the zero projection rays. However, by such simple method, the bottom side of the OS cannot be estimated from the truncated projection data. We verified that the top side of the OS alone is insufficient to get reconstruction comparable to **Fig. 6**(c). Moreover, such a simple method is not valid if the object is non-convex or the ROI is located completely inside the object (interior problem). A clinical example of this setup was studied here by modifying the phantom to simulate a female patient. In this setup, all the projection views suffer from the truncation in both the right and left portions. Even in this difficult setup, the solution is still unique if the OS is accurately known〔8〕. We simulated the same imaging setups as those of the previous studies and the results are presented in **Fig. 8**. Even with this challenging setup, the Th-OSEM behaves well and could reduce the artifacts in the reconstructed ROI.

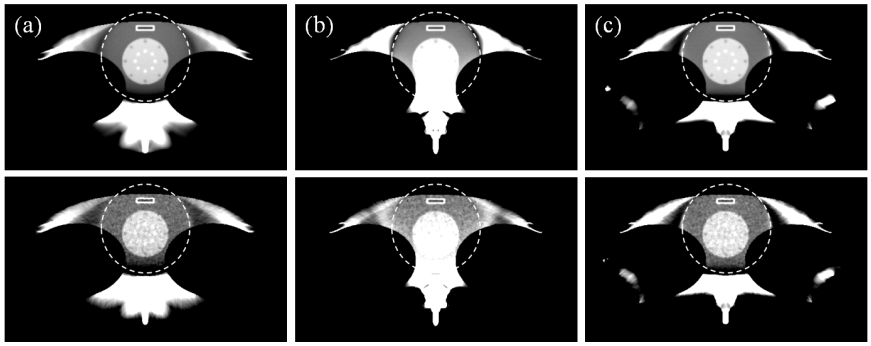

Fig. 6 Reconstructed ROIs by using (a) OSEM with the exact OS, (b) OSEM with the unknown OS and (c) Th-OSEM with the unknown OS. The top row shows reconstructions with noise free data and the bottom row shows reconstruction with added Poisson noise. The display gray scale is [0.95, 1.07].

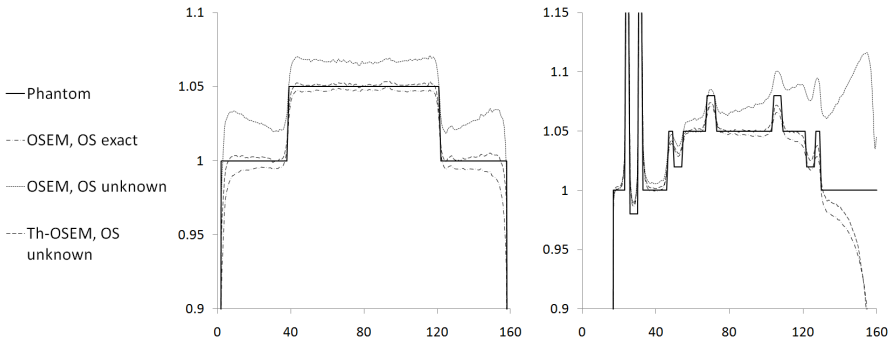

Fig. 7 Central horizontal (left) and vertical (right) profiles for ROI images shown in **Fig. 5** (top row).

In the simulation study to investigate the effect of OS accuracy on image quality (section 3.1), we assumed that the OS is known (with various degrees of accuracy) and used only the *a priori* knowledge that "the value of object is zero outside the known OS" (no *a priori* knowledge corresponding to the value of object inside the OS was used). On the other hand, in the simulation study to investigate the ability of the proposed algorithm (section 3.2), we assumed that the OS is unknown, because the purpose of this algorithm is to automatically estimate the OS. In other words, the proposed algorithm requires no *a priori* knowledge (except for the fact that the object has a finite support, which can be satisfied in every situation of CT imaging).

3）Real data

This study was done using a real data obtained from Siemens sensation 64 slice CT scanner using a 30×20×10 cm anthropomorphic chest phantom (QRM-thorax, QRM GmBH, Mohrendorf, Germany) shown in **Fig. 9**(d). The phantom consists of artificial lung lobes, shell of soft tissue equivalent material, tissue equivalent cardiac insert, and a spine insert, to which two water bottles were attached to represent the arms. Projection data was measured using the fan-beam geometry with 1,160 views (over 360º) and 672 detector channels. The projection data was rebinned into parallel-beam with 580 views (over 180º) and 672 radial bins. Image grid corresponding to the whole image was set to of 672×672 pixels and the ROI was selected as a circular region with a radius of 200 pixels covering the cardiac insert region. First, the FBP algorithm was used to reconstruct the whole object using the full scan data as shown in **Fig. 9**(a). Then a reconstruction was performed for ROI imaging using the OSEM and Th-OSEM

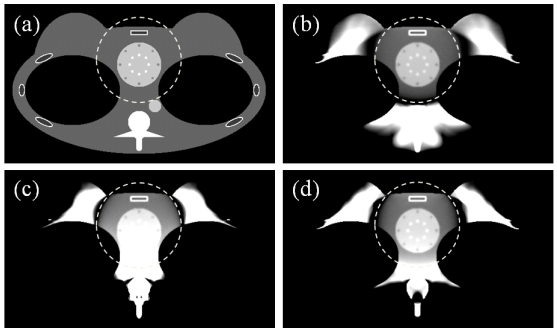

Fig. 8 (a) The modified phantom. Reconstructed ROIs using (b) OSEM with the exact OS, (c) OSEM with the unknown OS, and (d) Th-OSEM with the unknown OS. The display gray scale is [0.95 1.07].

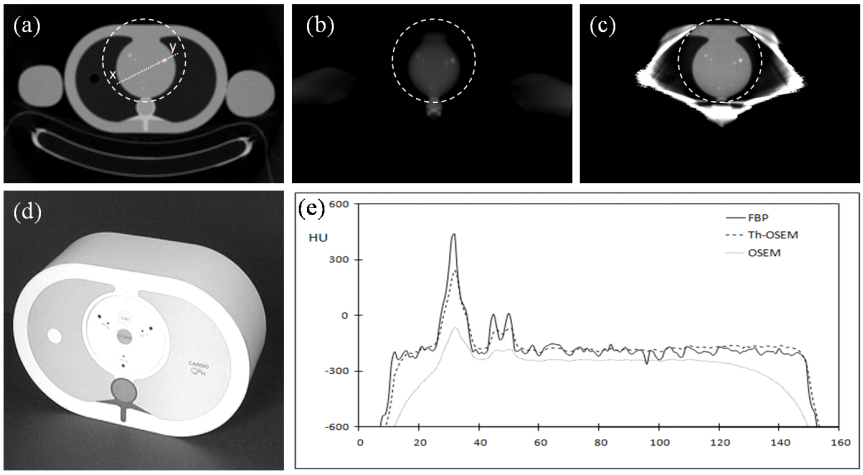

Fig. 9 Reconstructed images by using (a) FBP algorithm with the full scan, (b) OSEM with the ROI scan and (c) Th-OSEM with the ROI scan. The display window is set to [-480HU, 560HU]. (d) The original QRM-Thorax physical phantom (image from www.qrm.de). (e) Profiles corresponding to the white line segment (x-y) shown in (a).

algorithms with 16 subsets and 50 iterations without including the OS constraint. Reconstructed ROI images using OSEM and Th-OSEM are shown in **Fig. 9**(b) and (c), respectively. In **Fig. 9**, the reconstructed whole image matrix was trimmed to the ROI image consisting of 672×430 pixels for display. The results indicate a significant improvement in reducing the DC-shift and low-frequency artifacts when the Th-OSEM is employed.

4. Conclusions

The target of this work was to investigate the effect of OS constraint to the quality of reconstructed images from truncated projection data. This study implies that the lack of OS knowledge introduces DC-shift and low-frequency artifacts in the reconstructed images when the projection data is truncated. These artifacts decrease relative to the degree of OS accuracy. Moreover, we proposed an iterative algorithm that includes a sparsity constraint during the iteration in the form of thresholding function. This algorithm succeeded in reducing the artifacts thanks to its ability to automatically estimate the OS during reconstruction. The simulation results for noise free and noisy data show that the use of the proposed algorithm satisfactorily reduces the DC-shift and low-frequency artifacts even for the case where the OS is completely unknown. Moreover, the similar image improvements were achieved under more difficult configurations such as the non-convex object. The similar image improvements were also obtained by using the real data corresponding to the physical thorax phantom. The proposed algorithm is relatively simple and easy to be implemented and its extension to 3D reconstruction is also possible.

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物体境界未知下におけるトランケーションCT投影データからの逐次近似法による関心領域画像再構成

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要旨：本論文では，トランケーションされたCT投影データからの画像再構成を取り扱う．X線を関心領域（ROI）のみに照射してROI画像を撮影するROIイメージングでは，投影データがトランケーションされ画像再構成は解が一意に定まりにくい難しい問題となる．最近開発された微分逆投影（DBP）の理論によりROIのサイズと配置によってはトランケーションされた投影データから厳密な画像再構成が可能な場合があることが示されたが，物体境界が事前に既知であることが前提となっている．しかし，実際のイメージング状況では物体境界が正確にわからない場合が多い．本論文では，(1) トランケーションされた投影データからの画像再構成において事前情報として用いる物体境界の精度と再構成画像のアーティファクトの関係を評価すること，(2) 逐次近似画像再構成の反復計算の過程において物体境界を自動推定しアーティファクトを低減する新しい画像再構成法を提案すること，の2つを目的とする．新しい手法では，画像再構成のコスト関数に画像のL0ノルムを加えることで物体境界の推定を実現している．シミュレーション実験と実データによる実験を行い，提案手法を用いることによりアーティファクトが大幅に低減できることを示す．

キーワード：トモグラフィ，画像再構成，ROIイメージング，トランケーション投影データ，物体境界
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